

SECTION 21 12 00 - FIRE-SUPPRESSION STANDPIPES

PART 1 GENERAL

1.01 RELATED REQUIREMENTS

- A. Section 21 05 00 - Common Work Results for Fire Suppression: Fire protection piping.
- B. Section 21 05 23 - General-Duty Valves for Water-Based Fire-Suppression Piping.

1.02 REFERENCE STANDARDS

- A. NFPA 14 - Standard for the Installation of Standpipe and Hose Systems 2019.
- B. UL 405 - Fire Department Connection Devices Current Edition; Including All Revisions.

PART 2 PRODUCTS

2.01 FIRE DEPARTMENT CONNECTIONS

- A. Type: Free standing made of corrosion resistant metal complying with UL 405.
 - 1. Sleeve: Brass, 18 inches (460 mm) height.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Install in accordance with NFPA 14.
- C. Connect standpipe system to water source ahead of domestic water connection.
- D. Flush entire system of foreign matter.
- E. Buried piping and fittings:
 - 1. Cement-mortar lined, ductile iron pipe conforming to AWWA C-151, Class 200, 21/45 with Tyton or as approved, mechanical joints and fittings conforming to NFPA-24.
 - 2. Install pipe joints anchored with tie rods and restraints. Provide concrete thrust blocks at elbows and offsets conforming to NFPA-13 and NFPA-24.
 - 3. Other types of UL/FM reviewed piping including PVC C900 Class 200 may be used as approved by University's Representative and the State Fire Marshal.
 - 4. Thrust block installation shall be inspected by State Fire Marshal.
- F. Above ground piping and fittings:

1. Standard weight, Schedule 40, welded and seamless black steel, ASTM A-135 or A-53 pipe conforming to NFPA-13.
 2. Fittings for pipe sizes 2-1/2 inch and smaller shall be malleable iron, black, 150 lb. standard or cast iron, 175 lb. WP, screwed ends.
 3. Fittings for pipe sizes 3 inch and larger may be as above or weld fittings, ANSI 16.9, Schedule 40, flanged or welded fittings, cast iron, 175 lb. WP.
 4. Clamp type UL/FM approved fittings may be used on concealed piping where listed by the State Fire Marshal.
 5. Piping installed outdoors shall be galvanized.
- G. Alternate piping system to be Victaulic, Grinnel, or equal grooved piping systems.
- H. CPVC piping may be used in locations approved by NFPA 13 and the State Fire Marshal.

END OF SECTION 21 12 00